Geophys. J. Int. (2000) **140,** 357–373

Palaeomagnetism of the Miocene Farellones formation (Chile)

A. Goguitchaichvili,[1] A. Chauvin,[2] P. Roperch,[3] M. Prévot,[1] L. Aguirre[4] and M. Vergara[4]

[1] *Laboratoire de Géophysique et Tectonique, CNRS and Université de Montpellier* 2, 34095 *Montpellier, Cedex* 05, *France*
[2] *Géoscience Rennes, CNRS-UPR* 4661, *Campus de Beaulieu,* 35042 *Rennes, France. E-mail: Annick.Chauvin@univ-rennes1.fr*
[3] *IRD and Dep. Geologia, University of Chile, Santiago, Chile. E-mail:prperch@dgf.uchile.cl*
[4] *Dep. Geologia, University of Chile, Santiago, Chile*

Accepted 1999 September 21. Received 1999 September 13; in original form 1998 July 10

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

SUMMARY

We report on a detailed palaeomagnetic study of the Miocene Farellones volcanic formation in the Chilean Andes near Santiago (two sections, 37 sites, about 400 orientated cores). Petrological observations show evidence of low-grade metamorphism increasing downwards through the volcanic sequence. Optical observations of opaque minerals and magnetic experiments suggest that in many cases maghemitization is associated with hydrothermal alteration. However, thermal demagnetization data indicate that the low-grade metamorphism did not significantly modify the direction of the primary remanent magnetization recorded at the time of emplacement of the volcanic lava flows. Four intervals of polarity with two intermediate palaeodirections were observed in the $\sim$650-m-thick composite section. According to the dispersion of flow average directions, palaeosecular variation was slightly larger than that observed in general during the Upper Cenozoic. The site mean directions obtained in this study differ significantly from the expected Miocene direction. Clockwise rotations of up to 20° of small blocks are probably associated with the deformation of the Andean Cordillera since middle Miocene times. Geomagnetic palaeointensity data were obtained, using the Thellier method, on 24 samples from eight distinct lava flows. The flow mean VDM varies from 1.4 to 4.0×10^{22} A m^{-2}. Altogether, our data seem to suggest the existence of a relatively low geomagnetic field undergoing large fluctuations. Although a linear relationship was observed between the natural remanent magnetization and the thermal remanent magnetization acquired during the Thellier–Thellier experiments, undetected chemical alteration of the magnetic minerals during hydrothermalism may also explain the unusually low palaeointensity obtained.

Key words: magnetic mineralogy, Miocene, palaeointensity, palaeomagnetism, tectonic rotation.

INTRODUCTION

The time-averaged geomagnetic field is almost equivalent to the field produced by a geocentric axial dipole, a property which is at the basis of successful applications of palaeomagnetism to tectonics. The study of the time and space dependence of the secular variation of the field is important in understanding its origin. Camps & Prévot (1996) recently showed that a single statistical model of the geomagnetic field can describe the entire spectrum of direction and intensity fluctuations of the field from palaeosecular variation to reversal.

The geographical distribution of palaeodirection and palaeointensity data is still very uneven (e.g. Tanaka *et al.* 1995; Perrin & Shcherbakov 1997). In particular, very few data are available from the southern hemisphere, which hinders an accurate analysis of the fine-scale changes in the statistical characteristics of palaeosecular variations (Jacobs 1994). The main purpose of the present work is to improve our knowledge of the behaviour of the Earth's magnetic field in the southern hemisphere during the Miocene.

Many palaeomagnetic studies have been devoted to the study of the tectonics of the Chilean Andes [see Roperch & Carlier (1992) and Beck *et al.* (1994)]. Few data are available to describe secular variation and transitional field morphology (Brown *et al.* 1994). Up to now, there have been few determinations of palaeofield intensity from 16 to 20 Ma. The most detailed geomagnetic absolute intensity record for the Miocene is at the Steens Mountain (Prévot *et al.* 1985) transition

© 2000 RAS

(15.6 Ma). There is no determination from the southern hemisphere for the entire Miocene, except for the attempt by Derder *et al.* (1990) using volcanics from the Kerguelen Islands.

Although active volcanism related to subduction is widespread along the Andes, the nature of this volcanism provides few opportunities to observe thick volcanic sequences recording the evolution of the Earth's magnetic field. Such a sequence was found in the Chilean Andes near Santiago. In December 1995, we undertook a study of palaeomagnetic secular field fluctuations by sampling two sections (31 sites) of the Miocene Farellones volcanic formation. Additional palaeomagnetic sampling was performed in order to further constrain evidence of tectonic rotations that were found during the study of palaeosecular variation.

GEOLOGY AND SAMPLING

The Farellones Formation comprises the continental Miocene volcano-detrital sequence that forms a NS-orientated belt of outcrops, some 400 km long and 24–65 km wide (Vergara *et al.* 1988), between 31°30′S and 34°35′S along the central Chilean Andes, ~60 km east of Santiago. These deposits (average thickness ~1.5 km) are mainly located east of the Oligocene–Miocene magmatic belt and west of the present volcanic arc (Klohn 1960). The Farellones formation has a variable thickness along the outcrop belt. It is generally gently folded with dips not exceeding 15° (Charrier 1981, 1985).

The 'Los Farellones' formation was first defined by Munoz Cristi (in Hoffstetter *et al.* 1957) as a lower Tertiary continental volcanic sequence. Klohn (1960) suggested an Eocene age. Aguirre (1960) and Beccar *et al.* (1986) described this formation as a thick sequence composed of lavas, tuffs and brecccias of andesitic to rhyolitic composition with smaller quantities of dacites and basalts. A Miocene age was assigned to the Farellones formation by Thiele (1980) on the basis of preliminary radiometric data (Drake *et al.* 1976). According to additional K/Ar data (Vergara *et al.* 1988; Beccar *et al.* 1986; Munizaga & Vicente 1982), the volcanic activity occurred between 19.3 and 7.4 Ma. The ages of the largest volumes of extrusive rocks range from 18 to 15 Myr, thus defining a high intensity of volcanic activity during the early Miocene. The late Miocene–early Pliocene volcanic material (Colorado–La Parva Formation) is less abundant (Drake *et al.* 1976; Vergara *et al.* 1988). We sampled two volcanic sections belonging to the Farellones sequence (Figs 1 and 2). In general, 8–12 cores were drilled per volcanic unit. Cores were obtained with a gasoline-powered portable core drill, and then orientated using both a magnetic compass and sun orientation. The samples were distributed throughout each flow vertically and horizontally in order to check the geological continuity of the magnetic signal. Secondary magnetizations due to lightning seem to affect most of the natural outcrops located at high elevation or that form small peaks. However, most of our sampling sites, except for site VN17, correspond to outcrops from recent and deep road cuts.

A 150-m thick section (eight consecutive flows, VN28 to VN21 in Fig. 2) was sampled below the village of Farellones (33°21.21′S, 70°19.36′W, Figs 1 and 2), along the road to the ski resorts. A further stratified sequence (24 apparently consecutive lava flows, VN01 to VN20 and VN29 to VN31 in Fig. 2) was sampled below the ski resort of Valle Nevado

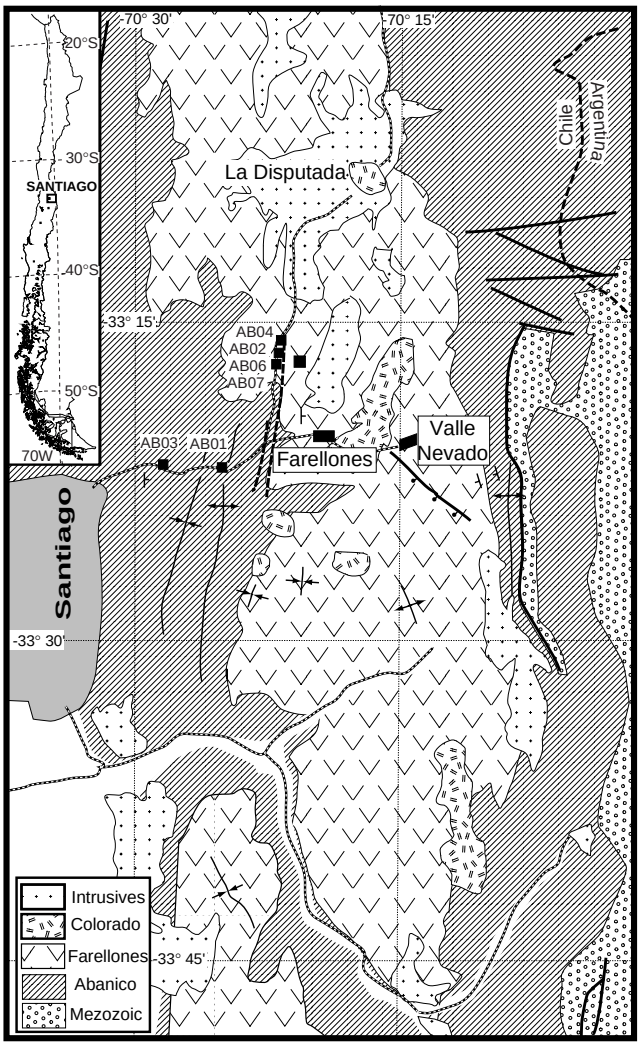


Figure 1. Location and geological setting of the studied area. Small squares indicate sampling sites in the Abanico units along the road to the mining centre (La Disputada). Two vertical sections were drilled along the road to the Farellones village and the ski resort Valle Nevado. The geological structures (faults and fold axis) have been shown according to the geological map of the area (Thiele 1980).

(33°21.48′S, 70°15.28′W). This 450-m thick sequence is located about 10 km east of and at a higher elevation than the 'Farellones' section. It is assumed to be younger and it is made of lava flows, flow-breccias, sills and dikes of basaltic composition, lahars, lapilli tuffs and subordinate unwelded pyroclastic flows. A 1-m thick subvertical dike cuts the lava flows and is observable at the individual flow levels from VN01 to VN10 (Fig. 2).

In many cases it was easy to identify the limits between successive flows. Flow bottoms sometimes present flattened and elongated amygdaloid vesicules and are altered, whereas the central parts seem more massive and, in most cases, free of alteration. Reddish oxidized fractures and veins are, however, observed at some sites. Flow tops show many fracture veins, filled mostly by various types of zeolites and clay minerals. Thin red layers of palaeosols were observed between VN02 and VN03, and the presence of sedimentary volcanic breccias may suggest a significant break in the volcanic activity between

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

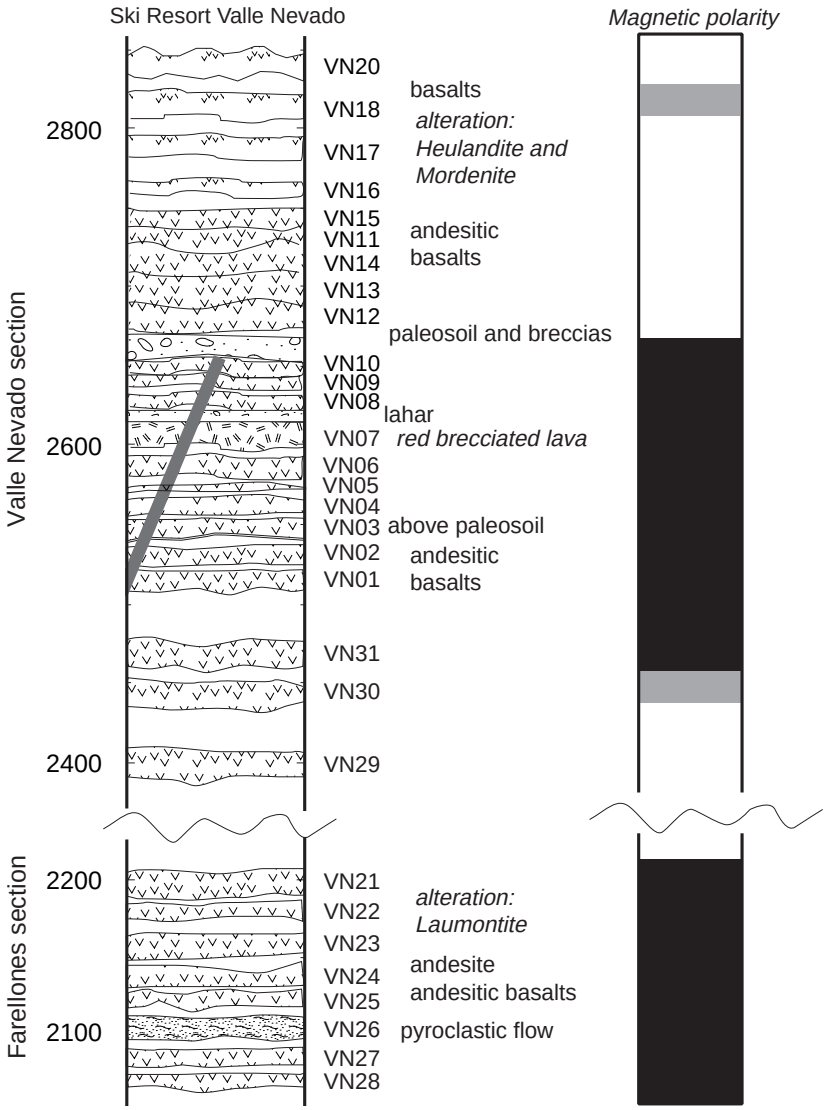


Figure 2. Synthetic column of the composite Farellones–Valle Nevado stratified sequence (VNS), and associated magnetic polarity column. The elevation is given in metres.

the emplacement of VN10 and that of VN12. A lahar is present between flows VN07 and VN08. Two of the volcanic units (VN26, VN07) correspond, respectively, to pyroclastic flow and brecciated lava flow, providing good markers within the stratigraphic columns. However, at a few places the stratigraphy was not obvious, and the sampling was later cross-checked in the field using the palaeomagnetic results. The contact between flow VN04 and flow VN05 was poorly defined. Three samples were taken from reddish altered lavas on top of unit VN05: owing to their distinct palaeomagnetic directions, it was decided to identify it as a separate unit. A few metres of poorly consolidated rocks were observed above this level, suggesting a possible gap in the volcanic activity.

Another short section, with only 3–5 samples per unit, was made about 500 m south of the main roadcuts at the same level as VN01 to VN11. Although these data confirm the main cross-section, we will not attempt to include these additional results because strong secondary magnetizations, mostly due to lightning, of the natural outcrops prevent the determination of an accurate mean direction for each unit. K/Ar analyses (Beccar *et al.* 1986) of fresh samples and plagioclase belonging to the upper members from near the La Parva volcanic centre of the Farellones formation (a few kilometres from the studied area) indicate age intervals of 16.6 ± 0.7 and 18.6 ± 0.6 Ma, respectively. A petrological study of the Valle Nevado sequence (VNS) units was performed recently by Aguirre *et al.* (1997). Microscopically, the lavas are porphyritic with intergranular, intersertal, pilotaxitic and felty types of groundmass where clinopyroxene, iron oxide, glass, and clay-minerals predominate. Olivines, in most cases, are completely altered to haematite/iddingsite, although in some flows fresh olivine crystals are preserved; Fe-Ti oxides are regularly abundant in all flows. Most chemical features of the magmatic products belonging to the VNS are similar to those found in volcanic arcs generated by subduction along continental margins (Aguirre *et al.* 1997). A very low-grade metamorphic process affected the rocks of the VNS. In the uppermost 100 m of the section (Fig. 2) the zeolites correspond to heulandite-clinoptilolite, while in the lowermost levels of the sequence laumontite is the main zeolite present. This low-temperature metamorphism seems to be related to neighbouring volcanic centres.

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

A more geographically distributed sampling was made on the volcanics (lavas, dikes and ignimbritic breccias) from the Middle Tertiary Abanico formation (Fig. 1). Four ignimbritic levels and one dike were drilled along the road to the mining district of La Disputada. Three volcanic units were sampled further west along the road from Santiago to Farellones. These sites are located to the west of our two main volcanic sequences, from which they are separated by a major fault. The Valle Nevado section does not show evidence for tectonic tilting or folding in our sampling area. A small tectonic correction (10°, to the east) was applied to the Farellones units (flows VN28 to VN20). Tectonic corrections were also applied to sites from the Abanico formation.

ROCK MAGNETIC PROPERTIES AND MAGNETIC MINERALOGY

Natural remanent magnetization and low-field susceptibility at room temperature were measured for all the samples. Most of the samples have susceptibility in the range 0.02–0.07 SI, while the intensity of the NRM ranges between 0.2 and 5 A m^{-1}. The highest NRM value was obtained from flow VN17, which was probably remagnetized by lightning (see below). The lowest NRMs are associated with the lowest susceptibilities and were observed for the volcanic units described as oxidized brecciated lavas or ignimbrite pyroclastic-type rocks. Within each flow, large variations in the magnetic properties can be observed whether the samples were drilled in the massive part of the flow or the brecciated and scoriaceous part of the lava flow. We will see later that samples from the within-flow brecciated levels usually have stable univectorial magnetizations with higher intensity of remanent magnetization and lower magnetic susceptibilities than samples from the massive part of the lavas.

High magnetic viscosity can prevent precise measurements of the remanence during progressive thermal demagnetization (Prévot *et al.* 1983). Xu *et al.* (1986) suggested that, in a wide variety of rocks types, viscous remanences can be acquired very rapidly, but can decay very much more slowly. The viscosity index v (Thellier & Thellier 1944) was determined on two samples per flow by measuring the remanent magnetization, first after a two-week storage with the ambient field parallel to the cylindrical axis of each specimen, and then after another two-week storage with specimens in the opposite direction. Most of these samples yield relatively low viscosity, varying between 0 and 10 per cent, except for flows VN09, VN11 and VN15 ($15 < v < 29$ per cent).

Susceptibility versus temperature

In order to identify the magnetic minerals carrying the remanence and to check their thermal stability, we made continuous measurements of susceptibility versus temperature (k–T curves) under a 10^{-4} mbar vacuum using a modified Bartington susceptibility meter MS2. One specimen per flow was heated to its Curie temperature with a heating and cooling rate of 10 °C min^{-1}. Additional experiments were made with the Kappabridge KLY3 at a heating rate of 8–35 °C min^{-1}. The Curie temperature was determined by the method of Prévot *et al.* (1983).

Three distinct types of behaviour were observed. Some of the curves show the presence of a single magnetic/ferrimagnetic phase with a Curie point compatible with relatively low-Ti titanomagnetite (flows VN02-03-07-09-10-12-14-15-20-21) (Fig. 3, sample VN2113C); however, the cooling and heating curves are not perfectly reversible. In other cases, two thermomagnetic phases could be distinguished during heating (flows VN01-03-06-11-13-14-16-23-24-25-28-29-31) (Fig. 3, sample VN2504A). The 'lowest Curie temperature' does not exceed 400 °C and can disappear near 400 °C (Fig. 3, sample VN2504A, curve 1). The highest ones range from 530 to 580 °C. Almost all cooling curves show only a single phase, with a Curie temperature close to that of magnetite. As experiments were carried out in a vacuum, the lowest Curie points found on such irreversible $k(T)$ curves could be due to the presence of titanomaghemites, which probably transformed into magnetite (Readman & O'Reilly 1972; Özdemir 1987) during heating. If this interpretation is correct, the susceptibility decrease observed between 300 and 400 °C might not correspond to a Curie point but to a dissociation temperature range. The low-temperature phase observed on both the heating and cooling curves of sample VN1307 (Fig. 3) may indicate Ti-rich titanomagnetites or intermediate titanohematites. Low-field thermomagnetic experiments on samples from brecciated and scoriaceous levels of the lavas reveal high Curie temperatures in the range 580–620 °C. This behaviour of low-field susceptibility is also correlated to high unblocking temperatures in a similar range of the natural remanent magnetization. An example is given by sample VN1813C (Fig. 3), which had one piece heated in a vacuum (solid line) and another in air (dashed line). The Curie point is apparently lower in a vacuum than in air but both are higher than 600 °C. Slightly maghemitized magnetite may have Curie temperatures of 610–645 °C (Özdemir & Banerjee 1984; Özdemir 1990).

We were not able to observe the Curie point of haematite on the k–T curves (under vacuum) for most weakly magnetized samples. Indications for haematite are a lower initial susceptibility associated with a lower NRM intensity and a large destructive field (MDF) of NRM. Flow VN26, for example, which yielded the lowest susceptibility and the lowest remanence, has a MDF larger than 200 mT, which could only correspond to haematite. This interpretation is supported by the isothermal remanent magnetization (IRM) acquisition curve, which indicates that saturation is not reached at 1.2 T.

Optical observations

The hydrothermal alteration suffered by the Farellones volcanics may have partially transformed the primary magnetic minerals and replaced the original TRM by pure CRM or a more complex magnetization. Identification of grain-growth CRM (which occurs at relatively low temperatures) is not easy. The size of the magnetic grains carrying the remanence is often too small and the petrological/optical observations of larger ferrimagnetic minerals may not completely correlate with the magnetic behaviour of the whole rock. Optical observations under reflected light were made on a few selected samples (12 samples taken from seven units) after palaeodirection and palaeointensity measurements.

In general, the studied samples present two populations of Fe-Ti oxides. The larger crystals (more than 30 μm) are rare, while the small crystals (3 μm and less) are abundant in all

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

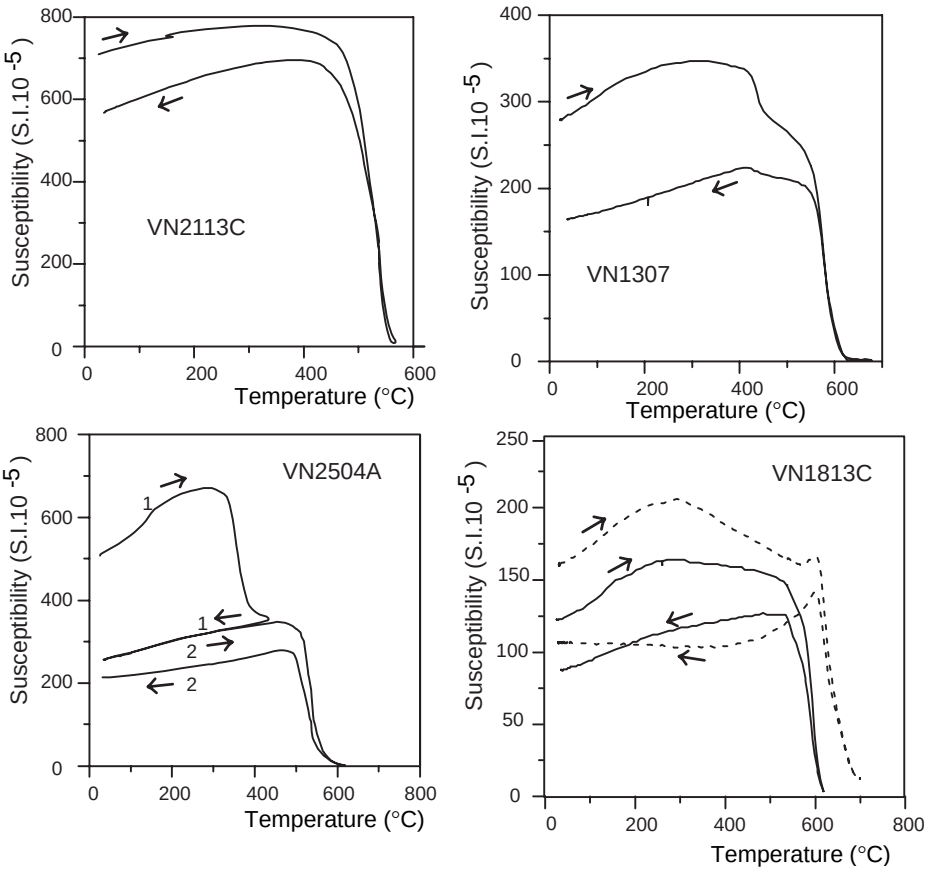


Figure 3. Susceptibility versus temperature curves of representative samples from Farellones formation units. The arrows indicate the heating and cooling curves.

studied samples. Both are irregularly distributed in the reddish matrix which is composed by plagioclase microlites, clays and oxyhydroxides of Fe. Sometimes the plagioclases are fractured and contain magnetic minerals. Two types of large opaques are observed. In most cases they are divided by several thin (less than 1 μm) grey, subparallel exsolution lamellae (Fig. 4a, sample VN1310C). The rest of the crystal shows quite a high reflectivity and strong anisotropy (Fig. 4b), which are characteristic of (titano)haematite. This haematite is finely polycrystalline in nature. In this context, the exsolutions that show no pleochroism or anisotropy are probably pseudobrookite (lamellar form, after Haggerty 1976). This paragenesis is characteristic of an extremely high state of oxidation. Similar observations were made for sample VN0705 (Fig. 5), but the exsolution lamellae are thicker. Microprobe analyses indicate that titanohaematite ($y\mathrm{FeTiO_3}$, $(1-y)\mathrm{Fe_2O_3}$) is poor in titanium ($y = 0.14$). The exsolutions observed in crystals from sample VN0705 correspond to an intermediate member of the ferropseudobrookite–pseudobrookite solid solution series ($v\mathrm{FeTi_2O_5} - (1-v)\mathrm{Fe_2TiO_5}$) with $v = 0.43$. However, the magnetic properties of this sample (see below) seem to correspond to some spinel rather than to haematite. Unfortunately, only macro-crystals are directly observable and we have no information about the small magnetic grains, which are abundant in this sample and which may carry most of the remanence. Evidence for titanohaematite-pseudobrookite intergrowths, which usually develop at the expense of ilmenite and titanomagnetite, is also observed for the flows VN12 and VN26. Most of the magnetic grains are very small (less than 1 μm). The extremely rare large crystals correspond to haematite associated with thin lamellae of pseudobrookite.

The second type of large opaques (Fig. 5b, sample VN1410) corresponds to the spinels without any exsolution. Neither anisotropy nor pleochroism was observed in these crystals. The crystal borders and the fractures are more reflective, indicating some alteration. Electron microprobe analyses indicate that x, the molecular percentage of ulvöspinel in the equivalent titanomagnetite (Prévot & Mergoil 1973), is about 0.24 in the apparently unaltered areas. Indeed, these areas seem to correspond to some highly oxidized titanomaghemite because the sum of oxides approaches 100 per cent only if it is assumed that the oxidation degree z is close to one. The most reflective altered part was at first believed to correspond to more maghemitized areas. Electron microprobe analyses do not confirm this view. It is now well established that maghemitization in nature results in an increase in x (Prévot *et al.* 1968; Petersen *et al.* 1979). In contrast, the more reflective areas of the titanomaghemite crystals were found here to be poorer in titanium ($x = 0.13$). This suggests that these altered zones correspond to a mixture of titanomaghemite and titanohaematite, the latter being due to a partial martitization of the former. Thus, mineralogical observations and microchemical analyses indicate that the original titanomagnetite was first transformed at low temperatures into a highly oxidized titanomaghemite. Then,

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

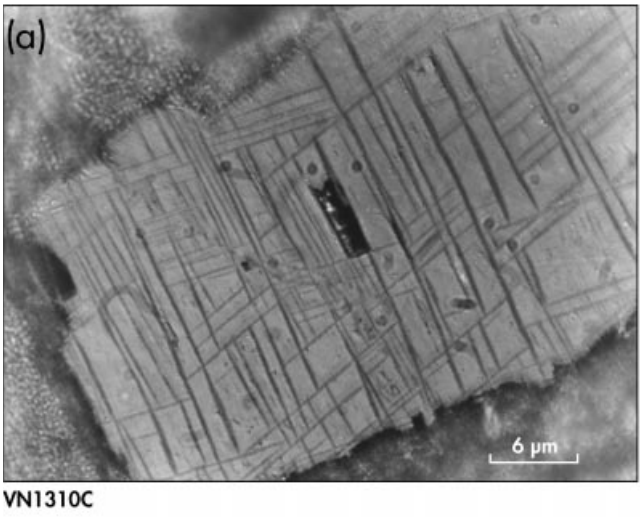


Figure 4. Typical example of a haematite-pseudobrookite assemblage (lamellar form): (a) under natural light; (b) under crossed nicols. Sample VN1310.

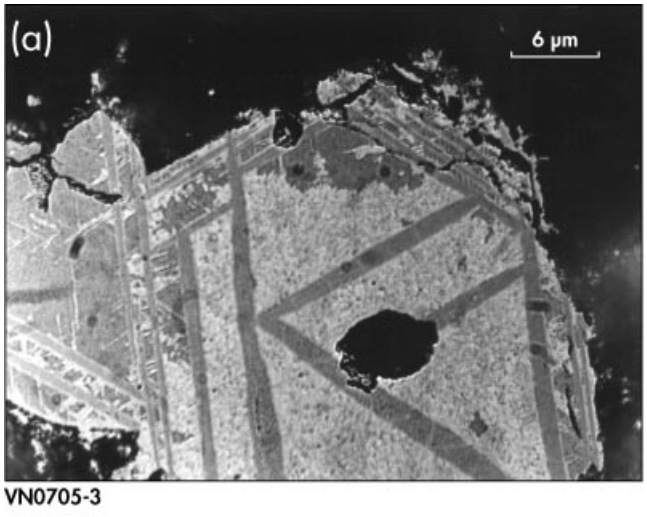


Figure 5. (a) An example of extremely high-degree oxidation resulting in the creation of haematite and pseudobrookite. Here, exsolution strips are thick. Sample VN0705. (b) Example of titanomaghemite. The reflectivity is often higher near fractures and crystal borders. Sample VN1410.

probably at higher temperatures, titanomaghemite began to transform into titanohaematite. These two alteration processes probably occur as a result of hydrothermalism (Merlet *et al.* 1993; Kelso *et al.* 1991). Similar observations were also obtained from flows VN04 and VN11. As pointed out above, most of the k–T curves are compatible with the presence of titanomaghemite.

It is worth noting that in a few cases (flows VN12 and VN14) some mineralogical inhomogeneity was observed within a given flow. Samples VN1203 and VN1410 yield evidence for titanomaghemite, while VN1209 and VN1407 apparently contain the haematite-pseudobrookite paragenesis described above. The possible effects of maghemitization and oxyexsolution on the remanence will be discussed below.

Rock magnetic analyses indicate a fairly complex mineralogy dominated by Ti-poor titanomagnetite and titanomaghemite with, in some cases, some titanohaematite. Mineralogical inhomogeneities are observed between the samples from brecciated and scoriaceous levels of the lavas and those from the massive parts. We will see later that the palaeomagnetic analysis strongly suggests that maghemitization did occur at the time of the emplacement of the volcanic unit.

PALAEODIRECTION DETERMINATION

Experimental procedure

Thermal stepwise cleaning in air or an alternating field (AF) was used to isolate the characteristic remanent magnetization using a Schonstedt furnace and AF demagnetizer. In some cases both kinds of cleaning were applied successively. At least one sample per core was continuously demagnetized. Remanent magnetization was measured in the Rennes laboratory using a Schonstedt spinner magnetometer. The magnetic susceptibility at room temperature was measured using a Bartington MS2 susceptibility meter, both before and after each step of thermal treatment in order to detect possible magnetic changes due to heating.

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

Quality of palaeomagnetic data

Most of the samples were demagnetized by stepwise heating up to 600–650 °C. Two distinct types of behaviour can be distinguished. In the first case, only one stable palaeomagnetic component is recognized (Fig. 6). For these samples, alternating field demagnetization (sample VN0108A) indicates a 20–25 mT median destructive field (MDF), which suggests that the remanence is probably carried by some fine-grained spinels. A higher-coercivity component (MDF = 35–40 mT) is also observed, but less frequently. Thermal cleaning shows that in several cases a significant fraction of the remanence is destroyed at temperatures above 580 °C (Fig. 6, samples VN1813 A and VN2211B). These observations are compatible with thermomagnetic experiments. Sample VN0705A comes from a red brecciated lava flow with magnetic properties compatible with a Ti-poor titanomagnetite. Indeed, a MDF value of 30–35 mT is obtained for this sample. Most of the unblocking temperatures lie around 500 °C (Fig. 4), which is common for Ti-poor titanomagnetite resulting from oxyexsolution.

In the other flows, two (or more) components of magnetization can be recognized (Fig. 7). Sometimes the first component is entirely destroyed below 250–300 °C (sample VN1210A). This component is probably a viscous remanent magnetization acquired during the last 0.78 Myr. Above 300 °C, reasonably good linearity is observed. In some cases the low unblocking temperature component (sample VN2109B) is not destroyed until 400–450 °C, suggesting that the origin of this magnetization is probably different. In some cases, it was obvious that the secondary overprint corresponds to a small isothermal remanent magnetization, as shown by AF demagnetization which was effective in removing the secondary magnetization.

One lava flow (VN17, 12 samples) was rejected for palaeodirection calculations because of evidence of remagnetization, probably due to lightning.

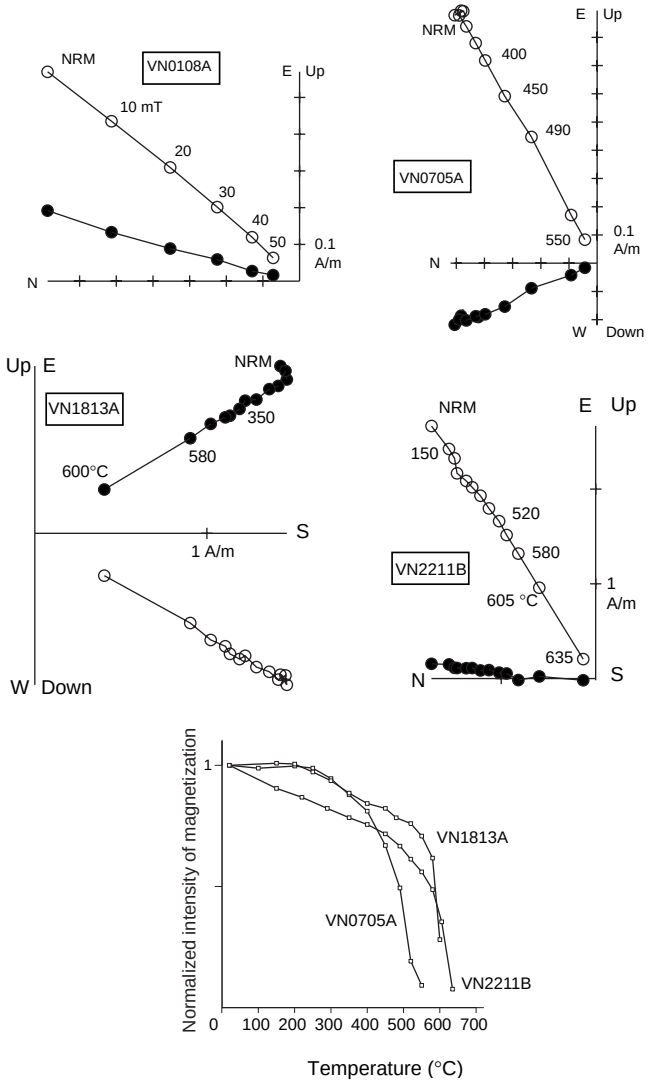


Figure 6. Orthogonal vector plots of stepwise alternating field demagnetization (samples VN0108A, VN2608C) and thermal demagnetization (samples VN0705A, VN1813A and VN2211B) of representative samples [solid (open) symbols: projection onto the horizontal (vertical) plane], and associated demagnetization curves.

Low-grade metamorphism and preservation of the primary remanent magnetizations

As discussed above, a very low-grade metamorphic process affected the rocks of the VNS, generating various metamorphic minerals which are found filling amygdales, vugs and veins. In most cases, we did not find strong evidence supporting the acquisition of late secondary magnetizations, or total or partial replacement of the primary magnetizations. We will discuss three distinct cases which illustrate an increasing degree of low-grade metamorphism.

The first case corresponds to samples pertaining to flows 11–15, for which detailed progressive thermal or AF demagnetizations reveal a similar characteristic direction (see discussion below). Fig. 8 shows the relationship between the intensity of NRM, susceptibility and NRM directions for all samples from these flows. Samples with the lowest susceptibility (10^{-2} SI) correspond to those having the largest NRM intensities (between 1 and 10 A m^{-1}). Moreover, these samples have NRM directions close to the characteristic directions determined for these flows. Secondary magnetizations are observed in samples with higher susceptibility values and NRM intensities lower than 1 A m^{-1}. We note that samples from the first group were drilled in the scoriaceous and autobrecciated levels of the lavas, while samples from the second group come from the massive parts of the lavas. We interpret this behaviour as being due to multidomain magnetite carriers within the massive parts of the lavas, while the initially oxidized levels have strong and stable magnetization acquired during initial cooling. Samples with the stable directions have high unblocking temperatures (median destructive temperatures around 570 °C), high MDF values (between 40 and 60 mT), and thermomagnetic experiments also show apparent high Curie temperatures above that expected for pure magnetite (Fig. 9). Various interpretations can be advanced to explain this magnetic behaviour. Slightly maghemitized titanium-poor titanomagnetite may be stable during heating and have high unblocking temperatures (Özdemir 1990). Alternatively, the presence of magnetite and titanohaematite might be responsible for the apparently stable magnetization.

The second case is illustrated by flow VN23. 10 samples from this flow were subjected to thermal demagnetization. Thermomagnetic analysis suggests the presence of maghemite

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

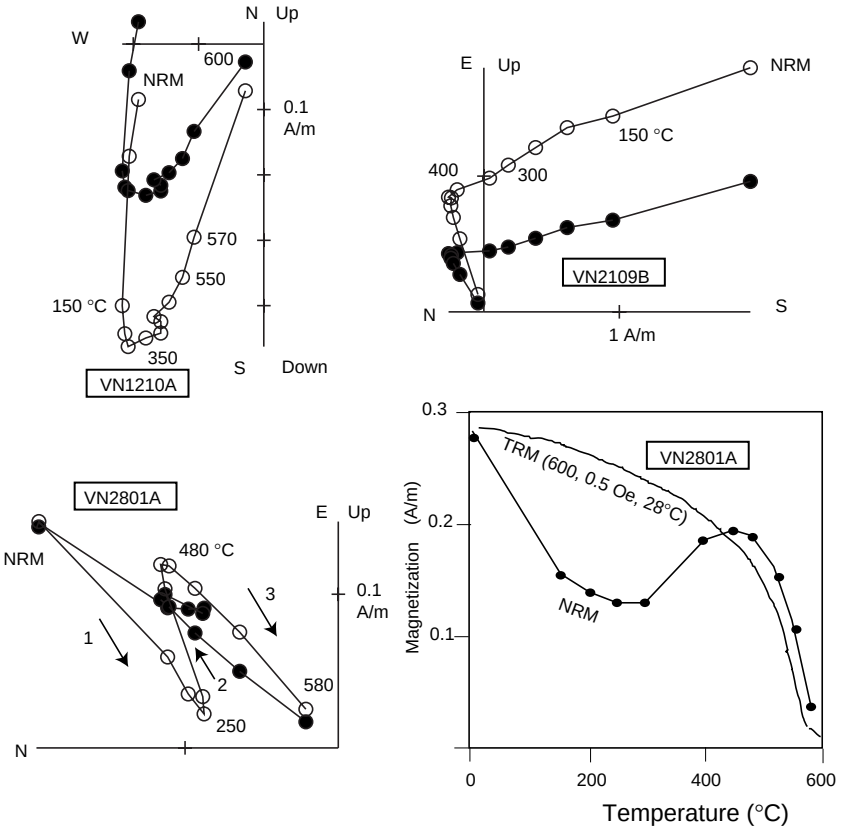


Figure 7. Orthogonal vector plots of stepwise thermal demagnetization from samples with two (or more) magnetization components. Same notation as in Fig. 6.

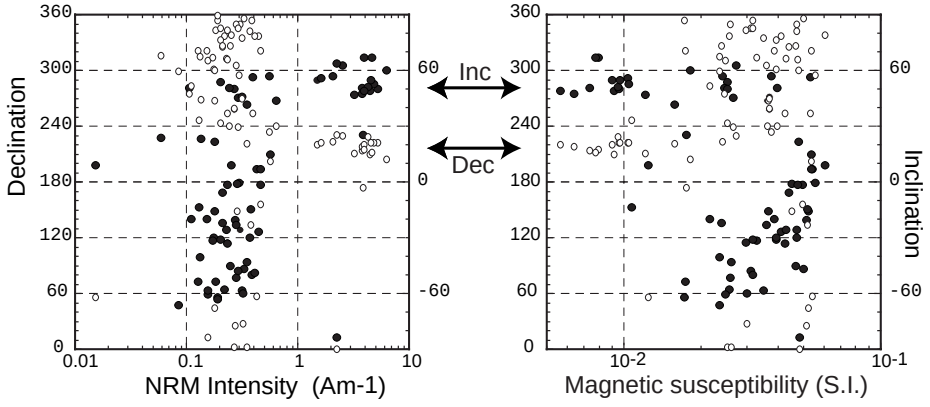


Figure 8. Relation between the NRM directions (declination: open circle; inclination: filled circle) and the intensity of NRM and magnetic susceptibility. All samples belong to a group of flows (VN11 to VN15) having similar reverse directions. The characteristic direction for these flows is indicated by the arrows.

in some samples (Fig. 10). Susceptibility data acquired after each step of demagnetization show in almost all samples a nearly 50 per cent decrease in susceptibility between 300 °C and 400 °C, suggesting the inversion of maghemite to titanomagnetite. In contrast, in the same temperature range, a large NRM intensity drop is observed only in some samples. This suggests that this magnetic phase does not carry a significant remanent magnetization. As can be seen in Fig. 10, there is no evidence for a distribution of the magnetic directions between two distinct directions. The primary magnetization is well defined.

The third case is illustrated by samples from flow VN28. Three components of magnetization are observed in some samples (Fig. 7, sample 2801A). The first component, between room temperature and 250 °C, yields a normal direction which is possibly a viscous overprint. Then, a component showing reverse polarity is unblocked from 250 to 480 °C. The last component of normal polarity, which is considered to be the characteristic magnetization, is demagnetized between 500 and 580 °C. Flow 28 provides us with unique evidence for the record of both polarities within the same flow. Such a behaviour can result from the superposition of a primary remanence with a

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

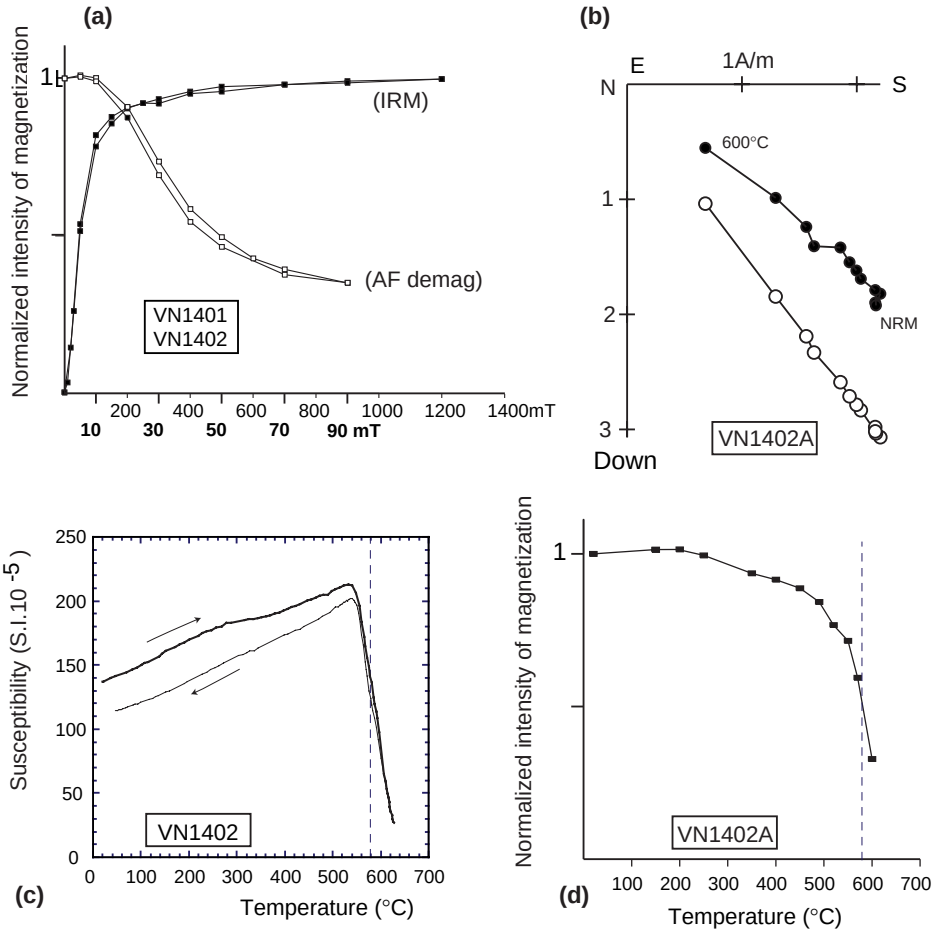


Figure 9. (a) Normalized intensity of NRM versus AF demagnetization for two samples, VN1401 and VN1402, showing a clear univectorial component and IRM acquisition curve for the same two samples. (b) Orthogonal plot of the thermal demagnetization of the NRM of sample VN1402. (c) Thermomagnetic experiment for sample VN1402. (d) Normalized intensity of NRM versus temperature during thermal demagnetization for sample VN1402. The dashed line at 580 °C is shown to highlight the observation of high unblocking temperatures significantly higher than the Curie point expected for magnetite.

CRM of opposite polarity acquired during hydrothermalism (Prévot *et al.* 1995). An alternative explanation might be some partial self-reversal of TRM. However, a continuous demagnetization of a total TRM (Fig. 7) acquired in the laboratory between 600 °C and room temperature in a 50 μT magnetic field does not show evidence of partial self-reversal. Flow 28 is located at the bottom of our volcanic sequence, where the low-grade metamorphic mineral assemblage suggests higher alteration.

Temperature estimates (150–200 °C) are derived from secondary clay minerals (Aguirre *et al.* 1997). We expected a stronger alteration of the remanent magnetization than the one observed. One possible explanation for the apparent discrepancy between the palaeomagnetic data and petrological observations is that a significant part of the hydrothermal alteration did occur at the time of emplacement of the lavas. Thus the chemical magnetizations associated with this process cannot be distinguished from the primary thermoremanent magnetizations. The second possible explanation is that hydrothermalism may result in some chemical changes (e.g. maghemitization), which might change the nature of the remanence without modifying its direction (Özdemir 1987). This explanation could be sustained if the characteristic remanent magnetizations with unblocking temperatures in the range 580–620 °C are indeed secondary CRM but with the same direction as the primary ones.

RESULTS

After thermal or AF stepwise demagnetization, a characteristic direction was determined by the least-squares method (Kirschvink 1980) for 82 per cent of the samples that were analysed. The directions obtained were averaged by flow, and the statistical parameters calculated assuming a Fisherian distribution. The average flow directions are precisely determined (Table 1 and Fig. 11). Almost all α_{95} are less than 11°, except for flow VN05 ($\alpha_{95} = 18°$) for which we had only three samples. The flows with an angular distance of more than 40° from the mean are regarded as intermediate.

Flows VN28 to VN20 are listed in stratigraphical order in Table 1, and the mean directions are given in Fig. 11. Only normal polarity flows are observed in the Farellones section (eight flows, $D = 34.6$, $I = -7.7$, $\alpha_{95} = 12.8$, $k = 20$). The Valle de Nevado section (Fig. 12) starts with a reversed polarity flow (VN29) followed by one intermediate polarity lava flow

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

Table 1. Palaeomagnetic parameters of the Farellones formation. *N*: number of specimens used for calculation, *I*: inclination, *D*: declination, *k* and α_{95}: precision parameter and radius of confidence cone of Fisher statistics, δD: angular distance from the overall mean direction of cleaned remanence of the flows, PLat/PLon: Latitude/Longitude of VGP position, *J*: geometric mean NRM intensity ($A\,m^{-1}$), χ: geometric mean susceptibility (10^{-2} SI units). Where tectonic tilting was observed, results are indicated both before (IS) and after (TC) bedding corrections.

Flow	*N*	*D*	*I*	α_{95}	*k*	δD	Plat	Plon	*J*	χ
VN20	16	187.7	21.5	5.3	99	34.8	−66.6	128.8	0.48	3.08
VN18	10	171.0	−4.3	4.8	103	64.3	−53.4	94.3	0.32	4.02
VN16	10	143.3	76.3	5.0	94	30.3	−52.1	314.7	0.61	3.80
VN15	8	223	47.5	5.3	109	14.9	−53.1	203.8	0.41	4.18
VN11	8	216.0	52.1	4.7	139	8.7	−60.0	208.1	0.37	3.38
VN14	10	224.9	50.8	8.9	31	14.3	−52.4	209.2	0.36	1.61
VN13	10	216.5	56.8	4.5	118	8.4	−60.2	217.5	0.66	2.24
VN12	10	223.3	51.8	4.2	136	13.1	−53.9	210.1	0.19	3.30
VN10	7	25.3	−48.3	9.9	37	6.6	68.1	15.3	2.67	4.20
VN09	7	3.3	−50.1	7.6	64	12.2	86.2	337.2	2.21	5.95
VNdike	8	15.4	−50.8	5.8	93	5.6	76.9	15.0	0.69	3.44
VN08	6	18.4	−50.3	6.1	124	4.8	74.3	15.6	1.62	3.01
VN07	7	339.8	−55.5	2.1	821	23.8	73.2	185.1	0.18	0.13
VN06	9	28.6	−37.9	5.9	76	17.3	61.9	1.0	1.76	2.81
VN05	3	266.4	39.3	18.1	47	44.8	−14.9	216.6	0.60	2.40
VN04	11	344.1	−47.2	3.9	140	24.7	75.4	215.9	2.15	3.56
VN03	22	295.0	−79.9	3.3	87	36.1	39.7	132.8	1.80	2.89
VN02	12	347.8	−37.5	4.3	103	28.6	73.5	245.6	1.17	2.21
VN01	15	18.1	−28.7	4.0	91	26.1	65.5	335.9	0.64	3.62
VN31	7	33.2	−58.1	4.9	156	7.1	62.8	40.3	3.03	2.04
VN30	7	134.8	−15.6	4.3	198	89.9	−30.4	55.0	0.33	1.60
VN29	6	125.3	71.3	4.8	197	35.4	−46.7	331.3	0.43	2.22
VN21 IS	10	7.9	−63.9	7.3	45	11.7	76.5	85.2	2.65	1.66
TC		26.3	−60.9				67.6	49.8		
VN22 IS	7	11.2	−51.9	5.6	119	7.0	80.6	16.7	1.01	3.72
TC		22.8	−49.0				70.3	15.7		
VN23 IS	10	70.5	−46.6	4.3	127	31.2	30.3	34.7	1.22	6.11
TC		73.3	−37.1				24.8	29.0		
VN24 IS	9	16.8	−60.5	4.5	134	6.5	74.5	55.3	0.70	2.29
TC		31.8	−56.3				63.9	36.4		
VN25 IS	4	46.3	−58.3	8.4	119	13.8	52.7	41.6	0.38	2.52
TC		55.2	−50.5				43.8	32.9		
VN26 IS	7	79.8	−68.9	5.0	145	29.4	31.8	64.5	0.06	0.03
TC		82.9	−59.0				25.5	51.9		
VN27 IS	7	37.9	−59.0	6.9	94	9.2	59.2	42.3	0.48	0.59
TC		48.6	−52.1				49.6	32.9		
VN28 IS	7	20.6	−32.7	2.4	617	21.9	65.8	344.2	0.53	1.52
TC		26.1	−28.7				60.3	348.6		
AB3a IS	9	207.2	50.9	6.9	56.3		−67.1	202.3	0.02	0.15
TC		213.9	41.9				−59.0	191.3		
AB3b IS	4	217.4	59.2	8.9	144		−59.6	222.8	0.47	3.34
TC		223.9	49.2				−52.8	206.8		
AB2 IS	7	355.1	−58.9	3.4	322		82.7	140.7	0.45	1.87
TC		354.3	−63.9				77.0	127.7		
AB1 IS	13	208.9	46.4	6.0	49		−64.5	194.1	0.35	2.07
TC		171.7	43.4				−79.2	65.6		
AB06 IS	8	345.2	−48.7	2.9	369		76.8	212.5		
TC		343.5	−53.5				76.3	192.2		
AB07 IS	7	15.2	−56.1	5.0	147		77.2	38.0		
TC		17.5	−60.9				73.8	56.5		

(VN30) and a normal polarity zone defined by 11 units, except for the anomalous direction of unit VN05. The reliability of this anomalous direction (observed on only three samples) is very low and this direction was discarded. The palaeodirection obtained from the dike has the same polarity as flows VN08, VN09 and VN10. This dike may be contemporaneous with this group of flows, since we have no geological or palaeomagnetic evidence that it cuts the whole sequence. The next five lava flows (VN12–VN15) show similar reversed directions (Fig. 12) and no major difference in petrography. Both these observations may indicate that these flows were emplaced during a short period of time. Between two flows carrying a reversed direction of magnetization (VN16 and VN20), a transitional direction is observed in flow VN18.

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

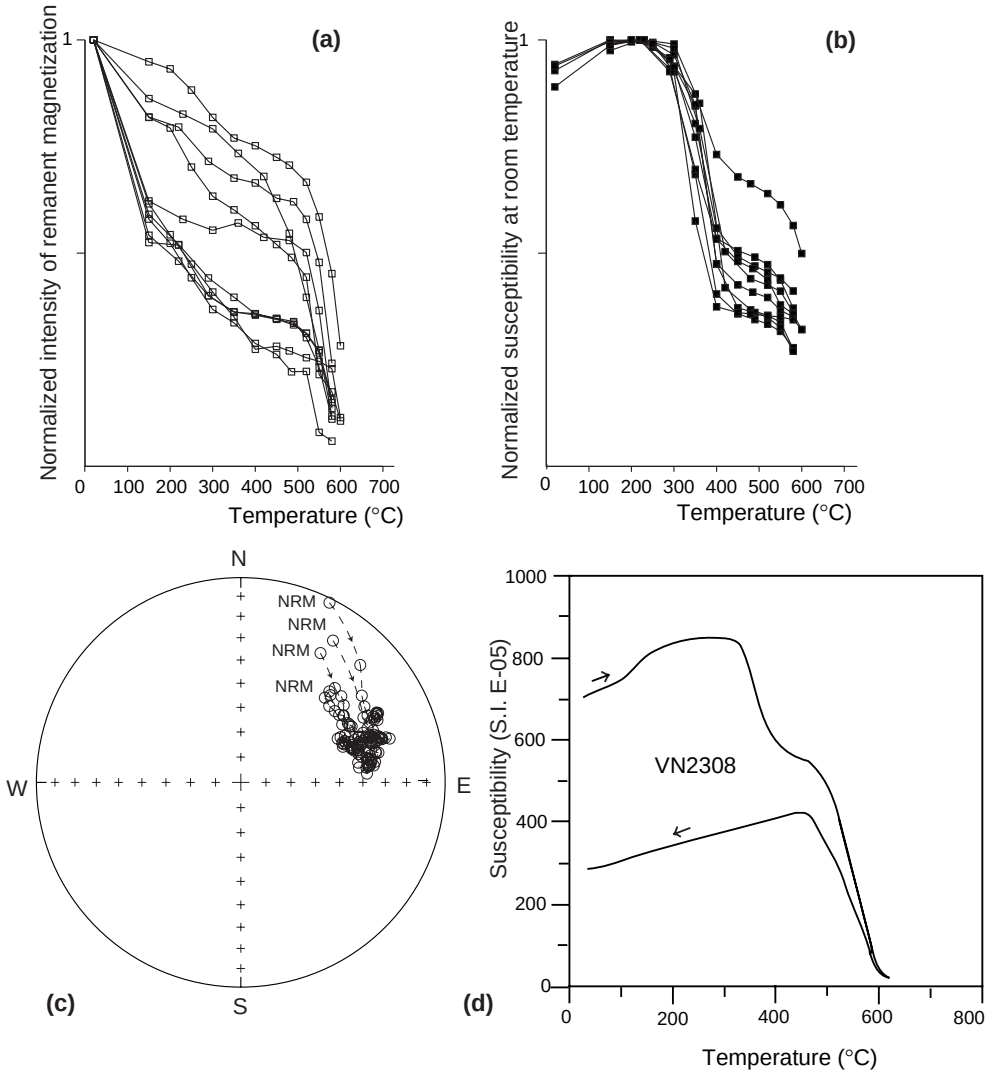


Figure 10. Palaeomagnetic behaviour of all samples from flow VN23 during thermal demagnetization. The sharp decrease in susceptibility between 300 and 400 °C (b) as well as the *k–T* curve (d) suggest that maghemite is one of the magnetic minerals in these samples. However, some samples show no decrease of the remanent magnetization in the same temperature range (a), suggesting that maghemite is not carrying a significant remanent magnetization. The directions are also well grouped (c).

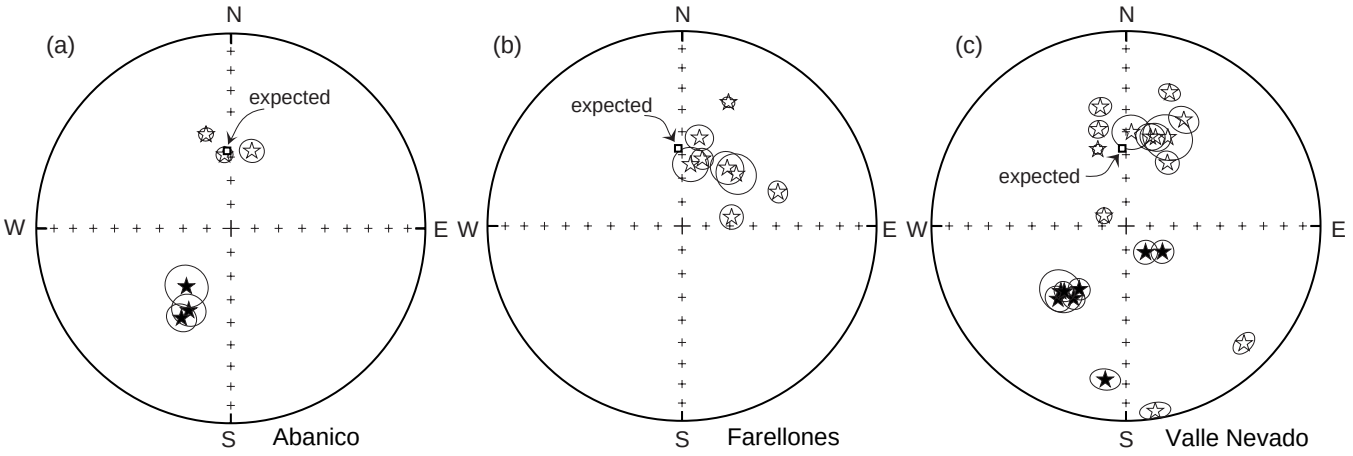


Figure 11. Equal-area projections of the characteristic directions of magnetization for the three groups of sites (a: Abanico; b: Farellones; c: Valle Nevado). Open (filled) symbols: negative (positive) inclination.

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

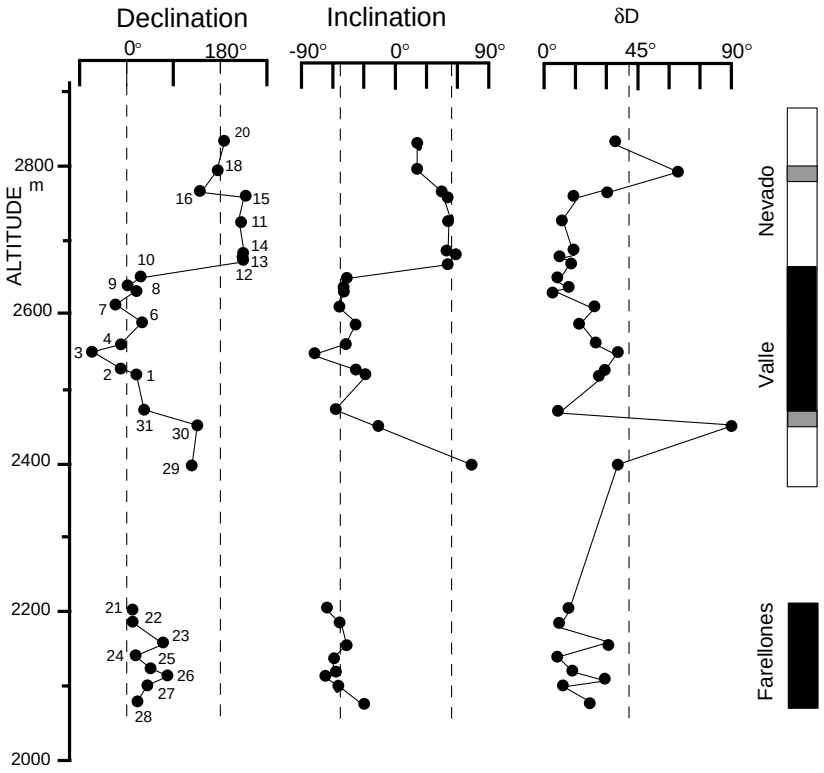


Figure 12. Evolution of various magnetic parameters for each flow. The dotted lines indicate the expected direction. The numbers refer to the flow numbers. δD is the angular distance from the overall mean direction of cleaned remanence of the flows (dotted line: cut-off angle used (40°)).

Normal and reversed directions of magnetization are observed at the six sites sampled in the Abanico formation (Table 1 and Fig. 9). No stable characteristic direction was obtained on one dike (Site AB05) and one volcanoclastic sediment (site AB04).

PALAEOINTENSITY EXPERIMENTS

Experimental procedure

The original version of the Thellier double-heating method (Thellier & Thellier 1959) was used. The specimens were heated twice (in the normal and reverse positions) at temperature steps successively increasing from room temperature to 600 °C (650 °C in some cases), with a minimum of 11 steps. All heatings were performed in a 10^{-2}–10^{-3} mbar atmosphere at the Montpellier laboratory under a 30 μT laboratory field which was held with a precision better than 0.15 μT. pTRM checks were performed every two steps in order to detect magnetic changes due to heating.

Sample selection

The Thellier method (1959) of geomagnetic absolute intensity determination, which is presumed to be the most reliable one, imposes many restrictions on the choice of samples that can be used (Coe 1967a,b; Levi 1977; Prévot *et al.* 1985; Pick & Tauxe 1993; Kosterov & Prévot 1998). In this study the sample selection was made according to three classic criteria: weak (less than 5 per cent) short-term viscosity index; the presence of only one stable magnetization component; and reasonable reversibility of the thermomagnetic curves. Owing to the presence of secondary components of magnetization or a complex magnetic mineralogy, only 52 samples belonging to 12 lava flows met our acceptance criteria.

Palaeointensity determinations

Only 24 samples belonging to eight flows yield palaeointensity determinations of good technical quality (Table 2). The NRM fraction f used for determination ranges between 0.34 and 0.93, and the quality factor q varies from 5.7 to 76. The palaeointensity values are reasonably coherent at the within-flow level for all flows studied except VN03, in which the within-flow dispersion exceeds 6 μT.

The highest-quality NRM–TRM diagrams obtained from this study are shown in Fig. 13. In some cases linearity is observed even up to 565 or 585 °C and the pTRM checks were successful. The lower-quality, but still acceptable, determinations correspond to samples for which the pTRM checks were significantly in error while the NRM–TRM curve is linear over more than one-third of the total TRM.

DISCUSSION

Average field direction and tectonic rotation

An informative palaeomagnetic record is obtained for the middle Miocene Farellones Formation. The Farellones section yields only normal polarity (Fig. 12), but three polarity zones

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

Table 2. Palaeointensity results from Farellones units. Flow is the flow name; Sample identifies the sample used for the palaeointensity determination; T_{min}–T_{max} is the temperature interval used for palaeointensity determination; n is the number of points defining a straight segment on the NRM–TRM plots; f, g and q are the fraction of extrapolated NRM used for intensity determination, the gap factor and quality factor (Coe 1978), respectively; Fa $\pm$ s are the individual palaeointensity estimate and associated error; Fe $\pm$ s and Fw $\pm$ s are flow mean and weighted flow mean palaeointensities, respectively, in μT; VDMe and VDMw are associated virtual dipole moments, in 10^{22} A m^2.

Flow	Sample	T_{min}–T_{max}	n	f	g	q	Fa $\pm$ s	Fe $\pm$ s	Fw $\pm$ s	VDMe	VDMw
VN20	2001B	20–565	12	0.48	0.79	11.4	15.4 $\pm$ 0.5	12.5 $\pm$ 3.8	13.2 $\pm$ 1.8	3.16	3.35
	2002B	20–565	12	0.34	0.82	6.7	9.6 $\pm$ 0.4				
VN14	1403B	192–584	11	0.54	0.79	27.2	23.1 $\pm$ 0.4	23.7 $\pm$ 1.1	24.1 $\pm$ 0.8	4.01	4.06
	1409B	243–620	12	0.86	0.81	76.1	24.3 $\pm$ 0.2				
VN13	1307B	352–584	10	0.93	0.68	20.6	22.1 $\pm$ 0.1	18.4 $\pm$ 4.2	17.8 $\pm$ 3.4	3.38	3.26
	1309C	352–584	10	0.87	0.75	42.6	17.3 $\pm$ 0.2				
	1310C	352–565	10	0.76	0.81	27.8	15.6 $\pm$ 0.3				
VN07	0702B	352–542	7	0.63	0.80	16.1	6.5 $\pm$ 0.2	7.7 $\pm$ 3.6	7.4 $\pm$ 1.7	1.42	1.36
	0703B	352–522	6	0.52	0.73	12.4	4.5 $\pm$ 0.1				
	0704A	352–565	7	0.51	0.78	15.4	5.7 $\pm$ 0.2				
	0705C	243–542	8	0.83	0.81	25.1	7.7 $\pm$ 0.2				
	0706C	192–542	10	0.42	0.85	11.5	13.8 $\pm$ 0.4				
VN03	0301C	192–522	10	0.41	0.86	16.7	9.6 $\pm$ 0.2	13.8 $\pm$ 4.1	12.5 $\pm$ 1.8	1.91	1.72
	0304C	243–565	13	0.64	0.85	36.4	9.2 $\pm$ 0.2				
	0310C	302–584	10	0.52	0.79	29.4	12.7 $\pm$ 0.3				
	0413B	302–584	10	0.52	0.84	16.1	15.2 $\pm$ 0.4				
	0416B	192–584	13	0.37	0.80	9.9	17.8 $\pm$ 0.5				
	0521A	352–565	8	0.36	0.79	5.7	18.7 $\pm$ 0.9				
VN02	0211B	402–565	7	0.79	0.72	15.6	9.3 $\pm$ 0.3	11.4 $\pm$ 2.4	11.2 $\pm$ 1.7	2.27	2.22
	0212B	243–522	8	0.63	0.68	12.3	13.1 $\pm$ 0.5				
VN31	3106B	192–584	12	0.37	0.73	11.1	16.5 $\pm$ 0.4	15.1 $\pm$ 2.1	14.8 $\pm$ 1.5	2.81	2.75
	3109B	302–584	9	0.61	0.81	15.8	13.6 $\pm$ 0.4				
VN24	2405C	243–620	11	0.78	0.76	27.1	10.5 $\pm$ 0.2	9.7 $\pm$ 1.2	10.2 $\pm$ 1.0	1.66	1.76
	2406C	302–584	9	0.62	0.72	11.2	9.3 $\pm$ 0.4				

are observed along the Valle Nevado section. Precise Ar^{40}–Ar^{39} radiometric ages are needed before attempting to correlate these magnetozones with the reference geomagnetic timescale.

Our palaeomagnetic data set has a fairly large dispersion. We calculated several mean directions (Table 3 and Fig. 14). In order not to include transitional directions, a cut-off angle was used to separate palaeosecular and intermediate field periods. McElhinny & McFadden (1997) used a VGP palaeolatitude of 45°, while Quidelleur *et al.* (1994) used 40°. Icelandic and low-latitude data indicate that the relative palaeointensity values (i.e. remanence intensities of basaltic rocks) decrease as the reversal angle increases, and stay very low for a reversal angle higher than 30°–40° (Kristjansson 1985; Chauvin *et al.* 1990). More recently, Camps & Prévot (1996) have shown that palaeosecular and transitional regimes can be distinguished for reversal angles of around 40°. This cut-off angle is used here, and two transitional directions are discarded (flows VN30 and VN18). The mean inclination is, in any case, in good agreement with the expected inclination ($I = -57°$) for the lower to middle Miocene period calculated from the reference poles given by Roperch & Carlier (1992). The declination is, however, significantly different from that

Table 3. Between-flow average directions and VGP for the Farellones volcanic units. FR: flows VN21 to VN28, VN: flows VN01 to VN20 and VN29 to VN31. Intermediate directions are discarded. Total represents all flows, including those coming from Abamico units, while in 'Total*' Abamico units are discarded.

Section	N	Inc	Dec	k	α_{95}	Plat	Plon	K	A_{95}
FR	8	−57.7	34.6	20	12.8	61.4	42.3	12	16.5
VN	19	−54.0	14.8	14	9.2	78.4	41.4	9	11.8
VN + FR normal	19	−54.7	18.2	16	8.5	74.5	38.5	10	11.1
VN + FR Reverse	8	57.1	205.6	12	17.0	−69.5	228.1	7	22.4
Total*	27	−55.5	20.3	15	7.4	73.1	41.9	9	9.6
AB Normal	3	−55.2	357.7	61	15.9	87.1	141.0	36	20.9
AB Reverse	3	52.2	210.7	126	11.0	−64.3	207.4	110	11.8
AB N + R	6	−54.9	14.9	37	11.1	77.6	33.8	21	14.9
Total	33	−55.4	19.3	17	6.2	74.0	40.7	10	8.1

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

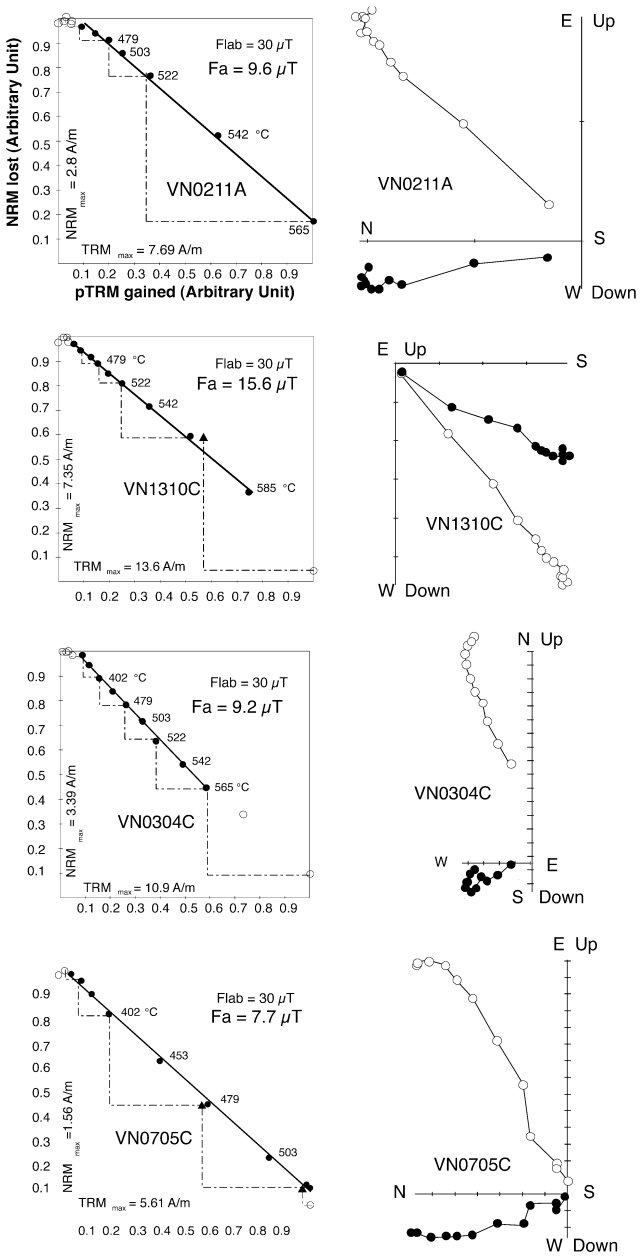


Figure 13. The highest-quality NRM–TRM plots and associated orthogonal diagrams from Farellones unit samples. In the NRM–TRM plots, triangles denote the pTRM checks. In the orthogonal diagrams we used the same notation as in Fig. 7.

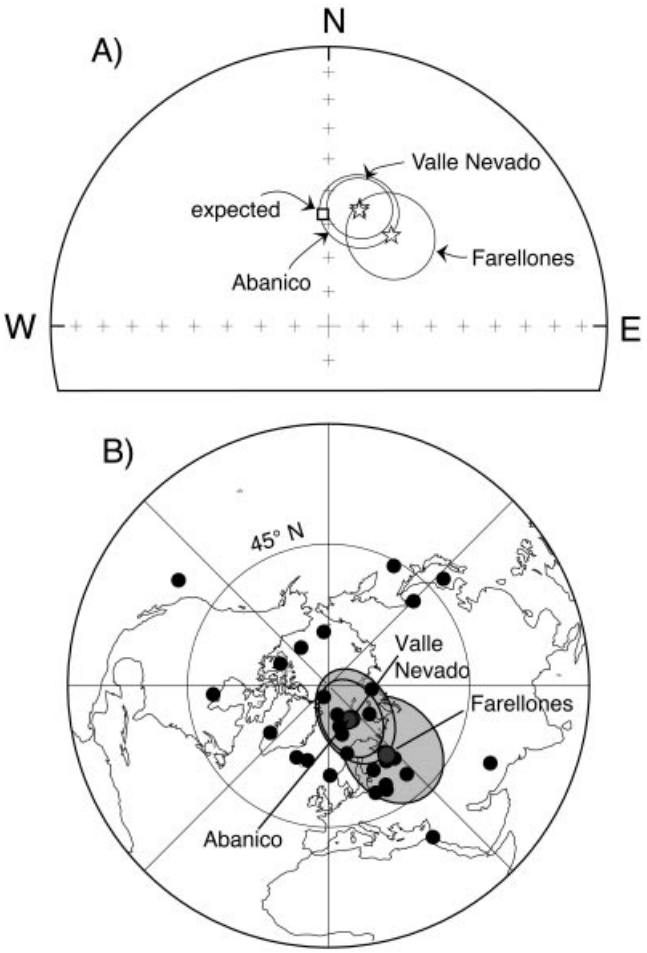


Figure 14. (a) The mean direction for the three groups of sites as indicated in Table 3. (b) Equal-area projection of the mean VGP per volcanic unit; transitional flows are discarded. The mean VGPs for the three groups of sites are in light grey.

expected ($D = 357^\circ$), except for the sites with normal polarity from the Abanico formation near La Disputada. The Farellones section shows the highest deviation in declination. The mean direction calculated using data from the Abanico and Farellones formations indicates a mean tectonic clockwise rotation of $23^\circ \pm 11^\circ$ (Demarest 1983).

Clockwise rotations in Central Chile have also been reported by Beck *et al.* (1994) in Cretaceous rocks. Our results in Miocene rocks suggest that rotations are associated with the deformation of the Precordillera. Further palaeomagnetic and structural work is needed in order to better describe the geodynamic evolution of the Precordillera of Central Chile.

Fluctuations of the palaeofield

Palaeomagnetic data in volcanic lavas provide spot-reading of the evolution of the geomagnetic field. Studies of the statistical dispersion of the palaeomagnetic directions are the only way to describe long-term variations. Such studies of the palaeosecular variations are numerous for the last 5 Myr. However, few data are available for the southern hemisphere and the Miocene period. We will discuss here the statistical dispersion of the Farellones and Valle Nevado data sets only, since the Abanico units may be slightly older than the middle Miocene units.

Owing to the evidence for clockwise rotations in the studied area, the dispersion was calculated around the mean direction or the mean VGP (after discarding two intermediate directions). The classical formula $S_F^2 = S_T^2 - S_W^2/n$ was used for estimating palaeosecular variation. Here, S_T is the total angular dispersion, S_W is the within-site dispersion, and n is the average number of samples per unit (Cox 1970). McFadden *et al.* (1991) report VGP scatter variations versus latitude for different time periods, suggesting that the magnitude of the secular variation is time-dependent. Their calculations are based on units whose VGP latitudes are more than 45° and indicate a VGP scatter

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

equal to 19 (upper and lower limits: 20.5 and 17.7) for the 33° latitude bands between 5 and 22.5 Ma. Using the same criteria, the S_F estimate obtained in this study, 21 (upper and lower limits: 25.8 and 17.7; Cox 1969), is slightly higher but supports McFadden *et al.*'s idea of significantly increased palaeosecular variation between 5 and 22 Ma.

Palaeointensity and possible effects of hydrothermalism

Absolute palaeointensity results (Fig. 15) show that even the maximum VDM value obtained in this study (4×10^{22} A m^2) is about half of the present-day one. There are few palaeointensities, determined using the Thellier method, available for the period 10–20 Ma. One determination was reported by Dunlop & Hale (1976) and two determinations using a total TRM method were given by Königsberger (1938). These results are very close to the present-day field intensity. However, the first study is based on submarine basalts, which are suspected to be maghemitized, and the second work is only of 'historical' interest. A significant number of data are available for the Steens Mountain sequence dated at 15.6 Ma (Prévot *et al.* 1985). They provide evidence for a relatively (30 per cent) reduced field before the reversal, but post-transitional flows provide VDM values similar to the present-day field. Nine Miocene high-quality palaeointensity estimates have been obtained on submarine basaltic glasses (Juárez *et al.* 1998). In this study, a virtual axial dipole moment (VADM) of 7.8×10^{22} A m^2 was obtained on the younger samples (6.4 Ma). On older samples (from 8 to 22 Ma), the VADMs determined are between 1.8 and 6.6, with an average of 3.6×10^{22} A m^2. Very recently, low palaeofield strengths have also been obtained on Miocene basalts and baked sediments from the French Massif Central (Riisager 1999). All these results seem to indicate that the Miocene dipole moment was significantly lower than the present-day one.

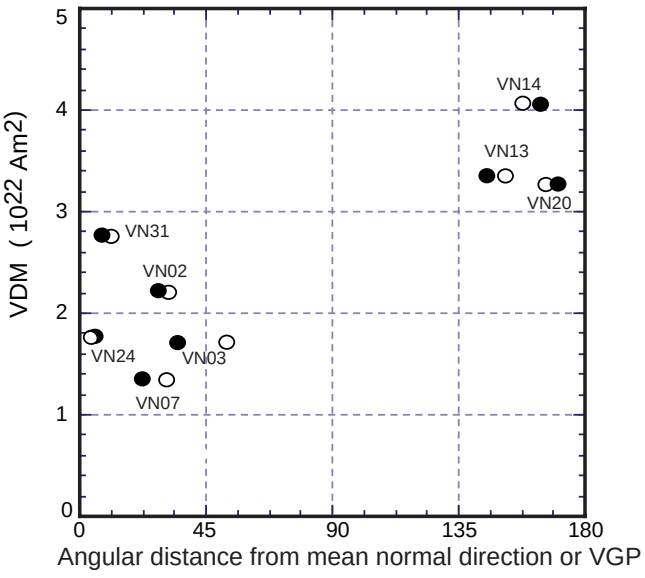


Figure 15. Virtual dipole moment (VDM) versus angular distance from the average mean direction (filled circles) or mean VGP (open circles).

Our results suggest the existence of a significantly weak field undergoing relatively large fluctuations in Chile during part of the Miocene. These observations could indicate a somewhat 'unstable regime' of the geomagnetic field around 17–20 Ma in the southern hemisphere. However, hydrothermalism could affect our palaeomagnetic data through some modification of oxide mineralogy and remanence characteristics.

Most of the flows studied seem to contain titanomaghemite, probably resulting from hydrothermalism, which might carry a chemical remanent magnetization. A crucial question is whether the CRM is acquired parallel to the ambient geomagnetic field during the low-temperature oxidation or is parallel to the primary TRM. Most of the experimental studies indicate that chemical magnetizations due to maghemitization of magnetite (Johnson & Merrill 1974; Heider & Dunlop 1987) or titanomagnetite (Özdemir & Dunlop 1985; Nishitani & Kono 1989) are recorded in the same directions as the original TRM. However, this might not to be the case for hydrothermalism (Kelso *et al.* 1991). In our case, we did not find strong evidence for an overprint due to late hydrothermalism. The between-flow palaeodirection changes are often large, while the within-flow characteristic directions are generally well defined; this suggests a correct recording of palaeosecular variation in direction.

In contrast, hydrothermalism can affect absolute intensity determination. McClelland (1996) suggested, on theoretical grounds, that grain-growth CRM carried by SD magnetite or haematite grains can be distinguished from a TRM on the basis of a Thellier palaeointensity experiment. In contrast to TRM, a grain-growth CRM should exhibit a concave-up Arai–Nagata plot, which is not observed here. Körner *et al.*'s (1998) recent experiments on samples in which a CRM was produced during long heating at a constant temperature showed that this theoretical prediction is not verified in practice: a near-linear Arai–Nagata plot is in fact observed. Furthermore, the apparent palaeointensity obtained from CRMs is always much lower (up to a factor of 2) than the applied field. Thus, even if the Farellones palaeointensity values are quite similar to other palaeointensities published for the same time period, we cannot reject the possibility that they could be large underestimates of the palaeofield strength.

CONCLUSIONS

The palaeomagnetic study of a 650-m thick sequence of andesitic-basaltic lavas suggests new evidence for relatively high palaeosecular variation of the geomagnetic field during the lower to middle Miocene. The higher dispersion of directions than usually observed for the last 5 Myr is perhaps correlated to a lower intensity of the geomagnetic field. Although petrological analysis indicates that low-temperature alteration is widespread throughout the sequence, it has not been possible to identify strong palaeomagnetic evidence of the record of secondary directions of magnetizations associated with late alteration. In contrast, the most stable magnetizations are found in the autobrecciated and scoriaceous levels, where alteration should be more important. The highly stable magnetizations may, however, correspond to secondary CRM with the same direction as the primary magnetization. The low-temperature oxidation which affects many flows raises some doubts regarding the significance of palaeointensity determinations.

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

The mean palaeomagnetic direction from the Farellones and the Abanico formations indicates that clockwise tectonic rotations may have occurred during the deformation of the Precordillera of Central Chile.

ACKNOWLEDGMENTS

This study was supported by CNRS-INSU and ORSTOM. (Contribution CNRS-INSU no. 165). AG's stay in Montpellier was possible through a PhD grant from the French Government.

REFERENCES

Aguirre, L., 1960. Geologia de Los Andes de Chile central, provincia de Aconcagua, *Instituto de investigationes geologicas,* Bol. no. **9**.

Aguirre, L.D., Robinson, D., Bevins, R.E., Fonceka, E., Vergara, M., Carrasco, J. & Morata, D., 1997. The Valle Nevado stratified sequence: Chemistry and alteration pattern, in *VIII Congreso Geologico Chileno, extended abstract,* pp. 1195–1199, Universidad Católica del Norte, Antofagasta, Chile.

Beccar, I., Vergara, M. & Munizaga, F., 1986. Edades K-Ar de la formation Farellones, en el cordon del Cerro la Parva, Cordillera de los Andes de Santiago, Chile, *Rev. Geol. Chile,* **28–29,** 109–113.

Beck, M.E.R., Burmester, R., Drake, R.E. & Riley, P., 1994. A tale of two continents: Some tectonic contrast between the Central Andes and the north American Cordillera, as illustrated by their palaeomagnetic signatures, *Tectonics,* **13,** 215–224.

Brown, L., Pickens, J. & Singer, B., 1994. Matuyama–Brunhes transition recorded in lava flows of the Chilean Andes: Evidence for dipolar fields during reversals, *Geology,* **22,** 299–302.

Camps, P. & Prévot, M., 1996. A statistical model of the fluctuations in the geomagnetic field from palaeosecular variation to reversal, *Science,* **273,** 776–779.

Charrier, R., 1981. Mesozoic and Cenozoic stratigraphy of the central Argentinian-Chilean Andes (32°-35°S) and chronology of their tectonic evolution, *Centralblatt Mineralogie, Geologie Paläontologie,* **1** (3/4), 344–355.

Charrier, R., 1985. Estratigrafia, evolution tectonica y significado de los Andes Chilenos entre 32 y 36°S, durante el Mesozoico y Cenozoico, in *Geologia y Recursos Minerales de Chile,* Vol. 1, pp. 101–133, Editorial de la universitad de Conception, Chile.

Chauvin, A., Roperch, P. & Duncan, R.A., 1990. Records of geomagnetic reversals from volcanic the islands of French Polynesia 2, Palaeomagnetic study of a flow sequence (1.2–0.6 Ma) from island of Tahiti and discussion of reversal models, *J. geophys. Res.,* **95,** 2727–2752.

Coe, R.S., 1967a. Paleointensities of the earth's magnetic field determined from Tertiary and Quaternary rocks, *J. geophys. Res.,* **72,** 3247–3262.

Coe, R.S., 1967b. The determination of the paleo-intensities of the Earth's magnetic field with emphasis on mechanisms which could cause non-ideal behavior in Thellier's method, *J. Geomag. Geoelectr.,* **19,** 157–179.

Coe, R.S., 1978. Geomagnetic paleointensity from radiocarbon-dated flows on Hawaii and the question of the Pacific nondipole low, *J. Geoph. Res.,* **83,** 1740–1756.

Cox, A., 1969. Confidence limits for the precision parameter k, *Geophys. J. R. astr. Soc.,* **18,** 545–549.

Cox, A., 1970. Latitude dependence of the angular dispersion of the geomagnetic field, *Geophys. J. R. astr. Soc.,* **20,** 253–269.

Demarest, H.H., 1983. Error analysis for the determination of tectonic rotation from palaeomagnetic data, *J. geophys. Res.,* **88,** 4321–4328.

Derder, M., Plessard, C. & Daly, L., 1990. Mise en évidence d'une transition de polarité du champ magnétique terrestre dans les basaltes miocénes des îles Kerguelen, *C. R. Acad. Sci. Paris,* **310,** Série 2, 1401–1407.

Drake, R.E., Curtis, G. & Vergara, M., 1976. Potassium-argon dating of igneous rocks in the Central Chilean Andes, *J. Volc. geotherm. Res.,* **3,** 285–295.

Dunlop, D.J. & Hale, C.J., 1976. A determination of palaeomagnetic field intensity using submarine basalts drilled near the Mid-Atlantic ridge, *J. geophys. Res.,* **81,** 4166–4174.

Haggerty, S.E., ed., 1976. Oxidation of opaque mineral oxides in basalts, in *Reviews on Mineralogy,* Vol. 3: *Oxide Minerals,* Mineralogical Society of America, Bookcrafters Inc., Chelsea, MI.

Heider, F. & Dunlop, D.J., 1987. Two types of chemical remanent magnetization during oxidation of magnetite, *Phys. Earth planet. Inter.,* **46,** 24–45.

Hoffstetter, R., Fuenzalida, H. & Cecioni, G., 1957. in *Lexique International, CNRS,* Vol. 5, Paris.

Jacobs, J.A., 1994. *Reversals of the Earth's Magnetic Field,* 2nd edn, Cambridge University Press, New York.

Johnson, H.P. & Merrill, R.T., 1974. Low-temperature oxidation of a single-domain magnetite, *J. geophys. Res.,* **79,** 5533–5534.

Juárez, M.T., Tauxe, L., Gee, J.S. & Pick, T., 1998. The intensity of the Earth's magnetic field over the past 160 million years, *Nature,* **394,** 878–881.

Kelso, P.R., Banerjee, S.K. & Worm, H.U., 1991. The effect of low-temperature hydrothermal alteration on the remanent magnetization of syntethic titanomagnetites: a case for acquisition of chemical remanent magnetization, *J. geophys. Res.,* **96,** 19 545–19 553.

Kirschvink, J.L., 1980. The least-squares method line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. astr. Soc.,* **62,** 699–718.

Klohn, C., 1960. Geologica de la Cordillera de los Andes de Chile Central, *Instituto de investigationes Geologicas, Santiago, Boll. 8.*

Königsberger, J.G., 1938. Natural residual magnetism of eruptive rocks, *Terr. Magn. Atmos. Elect.,* **43,** 299–320.

Körner, U., Prévot, M. & Poidras, T., 1998. CRM experiments and pseudo-paleointensity measurements on basaltic rocks with initially low Curie temperatures, EGS XIII General Assembly, Nice, *Ann. Geophys.,* 209.

Kosterov, A. & Prévot, M., 1998. Possible mechanisms causing failure of Thellier paleointensity experiments: results of rock-magnetic study of the Lesotho basalt, Southern Africa, *Geophys. J. Int.,* **134,** 554–572.

Kristjansson, L., 1985. Some statistical properties of palaeomagnetic directions in Icelandic lava flows, *Geophys. J. R. astr. Soc.,* **80,** 57–71.

Levi, S., 1977. The effect of magnetite particle size in paleointensity determination of the geomagnetic field, *Phys. Earth planet. Inter.,* **13,** 245–259.

McClelland, E., 1996. Theory of CRM acquired by grain growth and its implications for TRM discrimination and paleointensity determination in igneous rocks, *Geophys. J. Int.,* **126,** 271–280.

McElhinny, M.W. & McFadden, P.L., 1997. Palaeosecular variation over the past 5 Myr based on a new generalised database, *Geophys. J. Int.,* **131,** 240–252.

McFadden, P.L., Merrill, R.T., McElhinny, M.W. & Lee, S., 1991. Reversals of the Earth's magnetic field and temporal variations of the dynamo families, *J. geophys. Res.,* **96,** 3923–3933.

Merlet, C., Prévot, M. & Thomas, F., 1993. Determination of the oxidation degree of titanomaghemite in submarine basalts from direct analysis of oxygen with the electron probe microanalysis, *Terra Nova,* **5,** 88.

Munizaga, F. & Vicente, J.C., 1982. Acera de la zonation plutonica y del volcanismo Miocinico de Los Andes de Aconcagua, Datos radiometricos K/Ar, *Rev. Geol. Chile,* **16,** 3–21.

Nishitani, T. & Kono, M., 1989. Effects of low-temperature oxidation on the remanence properties of titanomagnetites, *J. Geomag. Geoelectr.,* **41,** 19–38.

Özdemir, Ö., 1987. Inversion of titanomaghemites, *Phys. Earth planet. Inter.,* **65,** 125–136.

Özdemir, Ö., 1990. High temperature hysteresis and thermoremanence of single-domain maghemite, *Phys. Earth planet. Inter.,* **65,** 125–136.

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026

Özdemir, Ö. & Banerjee, S.K., 1984. High temperature stability of maghemite, *Geophys. Res. Lett.,* **11,** 161–164.

Özdemir, Ö. & Dunlop, D.J., 1985. An experimental study of chemical remanent magnetization of synthetic monodomain titanomaghemite with initial thermoremanent magnetization, *Geophys. Res. Lett.,* **90,** 11 513–11 523.

Perrin, M. & Shcherbakov, V.P., 1997. Paleointensity of the earth magnetic field for the past 400 My: evidence for a dipole structure during the Mesozoic low, *J. Geomag. Geoelectr.,* **49,** 601–614.

Petersen, N., Eisenach, P. & Bleil, U., 1979. Low temperature alteration of the magnetic minerals in ocean floor basalts, in Deep Drilling Results in the Atlantic Ocean: Ocean Crust, *Am. geophys Un., Scientific Report,* **48,** 169–209.

Pick, T. & Tauxe, L., 1993. Geomagnetic paleointensities during the Cretaceous normal superchron measured using submarine basaltic glass, *Nature,* **366,** 238–242.

Prévot, M. & Mergoil, J., 1973. Titanomagnétites précoces et post-éruptives dans un basalte alcalin (Saint-Clément, Massif Central, France), in *Travaux Annuels du Département de Géologie et Minéralogie de Clérmont-Ferrand,* 142–174.

Prévot, M., Remond, G. & Caye, R., 1968. Etude de la transformation d'une titanomagnetite en titanomaghemite dans une roche volcanique, *Bull. Soc. Fr. Miner. Cristal.,* **91,** 65–74.

Prévot, M., Mankinen, E.A., Grommé, S. & Lecaille, A., 1983. High paleointensity of the geomagnetic field from thermomagnetic studies on rift valley pillow basalts from the middle Atlantic ridge, *J. geophys. Res.,* **88,** 2316–2326.

Prévot, M., Mankinen, E.A., Coe, R.S. & Grommé, S., 1985. The Steens Mountain (Oregon) geomagnetic polarity transition. 2, Field intensity variations and discussion of reversal models, *J. geophys. Res.,* **90,** 10 417–10 448.

Prévot, M., Thomas, F. & Perrin, M., 1995. Study of a remagnetization process affecting submarine basaltic lava flows, EGS XX General Assembly, Hambourg, *Ann. Geophys.,* 78.

Quidelleur, X., Valet, J.P., Courtillot, V. & Hulot, G., 1994. Long-term geometry of the geomagnetic field for the last 5 million years: an updated secular variation database, *Geophys. Res. Lett.,* **21,** 1639–1642.

Readman, P.W. & O'Reilly, W., 1972. Magnetic properties of oxidised (cation-deficient) titanomagnetites, (Fe,Ti,?) O_4, *J. Geomag. Geoelectr.,* **24,** 69–90.

Riisager, J., 1999. Variations en direction et en intensité du champ magnétique terrestre à la fin du Mésozoïque et au Cénozoïque, *PhD thesis,* Université de Montpellier.

Roperch, P. & Carlier, G., 1992. Palaeomagnetism of Mesozoic rocks from the Central Andes of Southern Peru: Importance of rotations in the development of the Bolivian orocline, *J. geophys. Res.,* **97,** 17 233–17 249.

Tanaka, H., Kono, M. & Ushimura, H., 1995. Some global features of paleointensity in geological time, *Geophys. J. Int.,* **120,** 97–102.

Thellier, E. & Thellier, O., 1944. Recherches géomagnétiques sur des coulées volcaniques d'Auvergne, *Ann. Geophys.,* **1,** 37–52.

Thellier, E. & Thellier, O., 1959. Sur l'intensité du champ magnétique terrestre dans le passé historique et géologique, *Ann. Géophys.,* **15,** 285–376.

Thiele, R., 1980. Hoja Santiago, Region Metropolitana, *Carta Geologica de Chile, no. 39,* Instituto de Investigaciones Geologicas, Santiago, Chile.

Vergara, M., Charrier, R., Munizaga, F., Rivano, S., Sepulveda, P., Thiele, R. & Drake, R., 1988. Miocene volcanism in the central Chilean Andes (31°30′S–34°35′S), *J. South Am. Earth Sci.,* **2,** 199–209.

Xu, T.C., Tarling, D.H., Eustance, N.B. & Hijab, B.R., 1986. Short-term viscous effects on measurements of remanence, *Geophys. J. R. astr. Soc.,* **87,** 305–309.

Downloaded from https://academic.oup.com/gji/article/140/2/357/781408 by Curtin University Library user on 29 April 2026